

MATHEMATICS – YEAR 8

Achievement Standard Guide



About this Achievement Standard Guide

This Achievement Standard Guide describes the expected knowledge, skills and understanding students should demonstrate for Mathematics in Year 8.

The Guides were developed using the achievement standards of Version 9.0 of the Australian Curriculum. The achievement standards assist teachers to monitor student learning and to make judgements about student progress and achievement.

The Guides set out the aspects of the achievement standard and describe indicators at five levels of attainment. They identify possible observable characteristics that could demonstrate the level of the student's understanding.

The Achievement Standard Guides were developed to:

- support teacher judgement when assessing student learning
- support planning for learning in all aspects of the achievement standard
- support the development of clear and specific learning intentions and success criteria.

The Achievement Standard Guides can be used to:

- ensure consistency of assessment
- support moderation of student work at year, school, network and state level
- support unit and lesson planning
- ensure teachers have sufficient evidence of learning to make defensible on-balance judgements about the quality of students' work against the achievement standards.

An effective assessment program will:

- Guide and support targeted teaching and learning
- Ensure students have opportunities to demonstrate depth and breadth of their learning in all aspects of the achievement standard
- Provide regular feedback to student about how they can improve in their learning
- Clarify future teaching and learning needs.

The following information supports teachers in these aspects of quality assessment practice.

Mathematics Year 8 Achievement Standard

Strand	By the end of Year 8, students:
<u>Number</u>	• recognise irrational numbers and terminating or recurring decimals • apply the exponent laws to calculations with numbers involving positive integer exponents • solve problems involving the 4 operations with integers and positive rational numbers • use mathematical modelling to solve practical problems involving ratios, percentages and rates in measurement and financial contexts
Algebra	• apply algebraic properties to rearrange, expand and factorise linear expressions • graph linear relations and solve linear equations with rational solutions and one-variable inequalities, graphically and algebraically • use mathematical modelling to solve problems using linear relations, interpreting and reviewing the model in context • make and test conjectures involving linear relations using digital tools
<u>Measurement</u>	• use appropriate metric units when solving measurement problems involving the perimeter and area of composite shapes, and volume of right prisms • use Pythagoras' theorem to solve measurement problems involving unknown lengths of right-angle triangles • use formulas to solve problems involving the area and circumference of circles • solve problems of duration involving 12- and 24-hour cycles across multiple time zones
Space	• use 3 dimensions to locate and describe position • identify conditions for congruency and similarity in shapes and create and test algorithms designed to test for congruency and similarity • apply the properties of quadrilaterals to solve problems.
Statistics	• conduct statistical investigations and explain the implications of obtaining data through sampling • analyse and describe the distribution of data • compare the variation in distributions of random samples of the same and different size from a given population with respect to shape, measures of central tendency and range
<u>Probability</u>	• represent the possible combinations of 2 events with tables and diagrams, and determine related probabilities to solve practical problems • conduct experiments and simulations using digital tools to determine related probabilities of compound events.

How the Achievement Standard Guide is organised

The relevant achievement standard statement in italics.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	Achievement standard statement with bolded variations For example, students can: • ...	Achievement standard statement For example, students can: • ...	Achievement standard statement with bolded variations For example, students can: • ...	

Content strand and sub-strand

'Can do' statements associated with each attachment level, including possible observable characteristics as examples of what students can do

Blank spaces or merged cells indicates a specific skill that can either be achieved or not.

Merged cells indicate that attainment is possible beyond 'At Standard' but not readily discernible between 'Above' or 'Well Above'.

Number

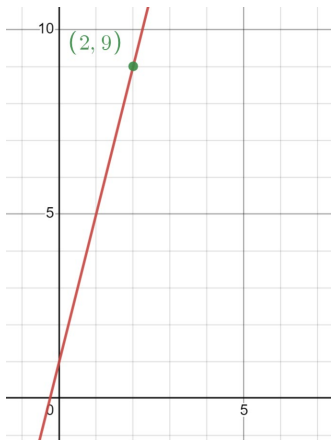
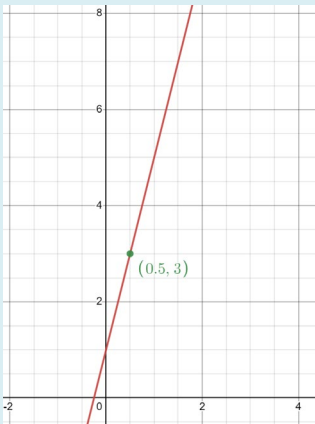
Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Number	With support, students recognise irrational numbers and terminating or recurring decimals. For example, students can: - Complete a sorting activity to differentiate irrational and rational numbers. - Describe the difference between recurring and terminating decimals.	Students recognise irrational numbers and terminating or recurring decimals. For example, students can: - Recognise irrational numbers (such as π) and can approximately locate them on a number line, between 2 terminating fractions. - Categorise decimals as terminating or recurring, using appropriate representations eg: a) $\frac{1}{3} = 0.\overline{3}$ b) $\frac{2}{11} = 0.\overline{18}$ c) $\frac{2}{15} = 0.1\overline{3}$ - Determine which fractions will have terminating decimal expansions or otherwise, based on the prime factorisation of the denominator, eg, without calculating, determine which of the following fractions will have a terminating decimal expansion: $\frac{5}{16}, \frac{7}{18}, \frac{13}{20}$	Students recognise irrational numbers and terminating or recurring decimals and convert terminating decimal expansions to fractions. For example, students can: - Apply prime factorisation to fractional representations of terminating decimals (to at least hundredths) and express in simplest form eg: $2.264 = 2 \frac{264}{1000} = 2 \frac{2^3 \times 3 \times 11}{2^3 \times 5^3} = 2 \frac{33}{125}$ - Explain and reason why some fractions will have terminating or recurring decimal expansions.	Students recognise irrational numbers and terminating or recurring decimals and convert terminating and recurring decimal expansions to fractions. For example, students can: - Convert from recurring decimals to fractions in simplest form and can clearly explain the process, eg: • $0.\overline{27}$ • $100 \times 0.\overline{27} = 27.\overline{27}$ • $100 \times 0.\overline{27} - 0.\overline{27} = 26.\overline{27} - 0.\overline{27}$ • $99 \times 0.\overline{27} = 27$ • $0.\overline{27} = \frac{27}{99} = \frac{3}{11}$

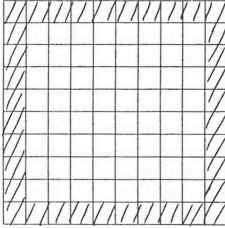
Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	With support, students apply the exponent laws to calculations with numbers involving positive integer exponents. For example, students can: Represent the following in expanded form: a) $2^3 \times 2^2$	Students apply the exponent laws to calculations with numbers involving positive integer exponents. For example, students can: Apply a single exponent (index) laws and combination of two laws to expressions involving the same base and positive integer exponents eg: a) $2^5 \times 2^3$ b) $\frac{3^7}{3^3}$ c) $\frac{2^5 \times 2^3}{2^2}$ d) $(3^4)^2$ e) $(2^5 \times 2^4)^3$	Students apply exponent laws, to calculations with numbers involving positive integer exponents, and different bases. For example, students can: - Apply a combination of two exponent laws to expressions involving different bases and positive integer exponents eg simplify the following: a) $(2^5 \times 5^8)^3$ b) $\frac{2^8 \times 2^3}{4^2}$ - Demonstrate for any non-zero number, n, that $n^0 = 1$, by for example recognising the pattern of dividing by 2, reduces the exponent of 2 by one each time, and/or by application of the second exponent law ie $\frac{2^7}{2^7} = 2^{7-7}$	Students apply combinations of exponent laws to calculations with numbers involving positive integer exponents and different bases, and connect the index laws with other number principles and laws to solve numerical problems. For example, students can: - Apply a combination of three or more exponent laws to expressions involving different bases and positive integer exponents: a) $\left(\frac{5^4 \times 3^7}{3^3 \times 5^3}\right)^4$ - Use prime factorisation and exponent laws to solve: a) $\frac{216}{64}$ - If $a = 2^5 \times 3^8 \times 5^3$ - And $b = 2^4 \times 3^8 \times 5^3$, find: a) $a + b$ b) $a - b$ c) $a \times b$ d) $a \div b$ expressing all answers in index form.
	With support, students solve problems involving the 4 operations with integers and positive rational numbers. For example, students can: Solve one step problems involving the 4 operations with integers and positive unit fractions and decimals to hundredths.	Students solve problems involving the 4 operations with integers and positive rational numbers. For example, students can: Solve one-step problems involving the four operations with integers and with positive fractions, decimals, and percentages.	Students solve multi-step problems involving the 4 operations with integers and positive rational numbers and apply the order of operations. For example, students can: Apply a range of strategies including pattern recognition, partitioning, regrouping via the commutative or associative properties, place value and so on, in multi-step problems.	Students solve multi-step problems involving the 4 operations with integers and positive rational numbers, apply the order of operations, and compare and evaluate different strategies. For example, students can: - Compare and evaluate different ways to solve 62% of 25×2. - What might be the most effective strategy and why?

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Algebra

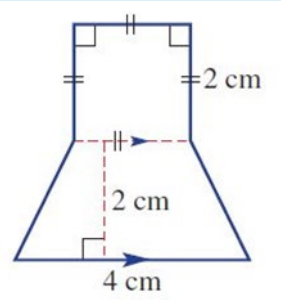
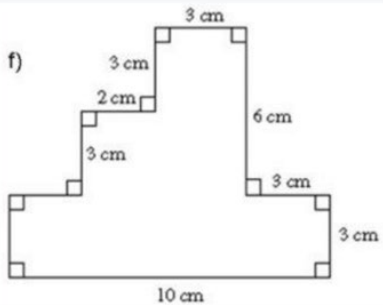
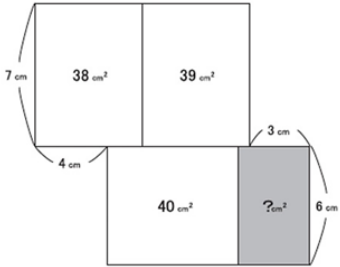
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Algebra	With support, students apply algebraic properties to rearrange, expand and factorise linear expressions. For example, students can: With scaffolding, expand, rearrange and factorise expressions. $5(j + 3) = 5 \times \square + 5 \times \square$ $= \underline{\hspace{2cm}}$	Students apply algebraic properties to rearrange, expand and factorise linear expressions. For example, students can: - Simplify straightforward expressions with different variables and constant terms. - Using distributive, associative, commutative, identity and inverse properties to expand and factorise straightforward expressions using concrete materials or the area model to support calculations and make connections between the two processes. - Simplify by collecting like terms: $3a + 2b - a + 3b + 7$ - Expand $3(x + 2)$ - Factorise $9y - 12$	Students apply algebraic properties to rearrange, expand and factorise complex linear expressions involving multiple steps or compound factors. For example, students can: - Expand and simplify expressions combining brackets and separate terms including negative terms and factors. - Factorise expressions that involves multiple variables and compound factors eg expand and simplify: $-3(2x + 5) - 2x$ - Factorise $24xy - 16x$	Students apply algebraic properties to rearrange, expand and factorise complex linear expressions involving multiple steps and compound factors. Factorise expressions with terms that include exponents. For example, students can: - Expand and simplify expressions with multiple sets of brackets. - Factorise expressions that involves terms with powers and multiple variables. Note that this does NOT include quadratic factorisation. - Expand and simplify: $x(2 - y) - 5(x + 2y)$ - Factorise $-9x^3y - 12x^2y$
	With support, students graph linear relations and solve linear equations with rational solutions and one-variable inequalities, graphically and algebraically. For example, students can: - Solve linear relations with integer solutions. - Graphs linear relations and makes connections between graphical and algebraic solutions eg:	Students graph linear relations and solve linear equations with rational solutions and one-variable inequalities, graphically and algebraically. For example, students can: - Solve linear relations and inequalities with rational solutions. - Graph linear relations and make connections between graphical and algebraic solutions eg:	Students graph linear relations and solve linear equations with rational solutions and one-variable inequalities with variables on both sides of the relation symbols, graphically and algebraically. For example, students can: Graph related linear relations and inequalities that include variables on both sides of the relation symbols (equals sign and inequalities), and makes connections between graphical and algebraic solutions	Students graph linear relations and solve linear equations with rational solutions and one-variable inequalities with variables and brackets on both sides of the equals sign, graphically and algebraically. For example, students can: Solves linear equations and inequalities involving brackets, and variables on both sides of the relation symbols and make

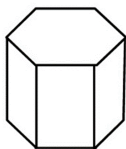
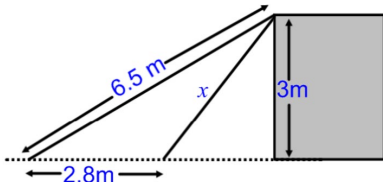
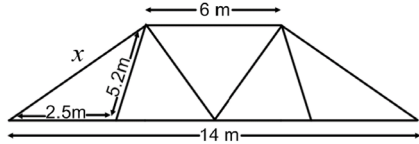
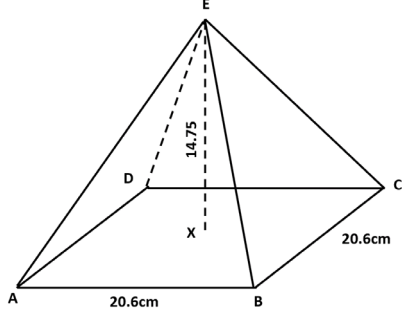
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	- Graph the relation $y = 4x + 1$ - Use your graph to find the value of x when $y = 9$. 	- Consider the relation $y = 4x + 1$, where $y = 3$. - determine the value of x graphically and confirms the answer using substitution.  - Graphically demonstrates solutions to related inequalities, such as $y > 4x + 1$.	eg solve the following linear equation algebraically: a) $5x + 10 = 3x$ and verify by substitution - On the same axes, plot $y = 3x$ and $y = 5x + 10$ and relate the algebraic and graphical solutions. - Demonstrate and describe the solution of related inequalities, such as $5(x + 2) \leq 3x$.	connections between graphical and algebraic solutions eg solve the following linear equation: $-5(x + 2) = 3(x - 2)$ and verify by substitution - On the same axes, graph $y = 3(x - 2)$ and $y = -5(x + 2)$ and describe the relationship between the graphical and algebraic solutions. - Demonstrate and explain the solution of related inequalities, such as $-5(x + 2) \geq 3(x - 2)$.
	With support, students use mathematical modelling to solve problems using linear relations, interpreting and reviewing the model in context. For example, students can: - Use a mathematical modelling approach that allows them to solve problems using linear relations by: a) translating the problem into maths b) using maths to solve the problem c) interpreting and communicating the solution	Students use mathematical modelling to solve problems using linear relations, interpreting and reviewing the model in context. For example, students can: - Use a mathematical modelling approach that allows them to solve problems using linear relations by: a) translating the problem into maths b) using maths to solve the problem c) interpreting and communicating the solution back in the real-world context d) justification of the mathematical representations	Students use mathematical modelling to solve problems using linear relations, interpreting, reviewing and extending the model in context. For example, students can: - Use a mathematical modelling approach that allows them to solve problems using linear relations by: a) translating the problem into maths b) using maths to solve the problem c) interpreting and communicating the solution back in the real-world context d) justification of the mathematical representations	Students use mathematical modelling to solve problems using linear relations, interpreting, reviewing and generalising the model in context. For example, students can: - Use a mathematical modelling approach that allows them to solve problems using linear relations by: a) translating the problem into maths b) using maths to solve the problem c) interpreting and communicating the

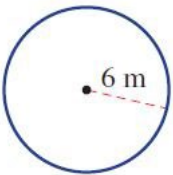
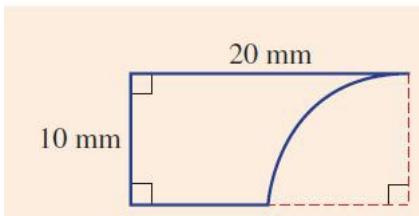
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	back in the real-world context d) justification of the mathematical representations e) interpreting and reviewing the model in context. - Use a given mathematical model to solve for specific unknown values, by completing a table of values or reading from a graph. - Eg a model given for high tide change over time is $H = 0.5d + 7$, where d is the day of the month and H is the number of hours that high tide occurs after midday. - Draw a graph of the model. - What value of H is given on the day 6 of the month?	e) interpreting and reviewing the model in context. - Select and justify the appropriate representations (word form, table, graph, diagram or equation) for the context of the problem. - Eg model given for high tide change over time is $H = 0.5d + 7$, where d is the day of the month and H is the number of hours that high tide occurs after midday. - Choose and present an appropriate representation for this model in the context of the questions below. - What value of H is given on the day 6 of the month? - On what day of the month will high tide be 9 hours after midday? - On what day of the month will high tide be 6 hours after midday? What might this value mean on your graph? - What would be appropriate values of d to use with this model? Justify your response. <u>Border problem:</u>  - Finds an algebraic solution for any $n \times n$ square.	e) interpreting, reviewing, and extending the model in context. - Works flexibly with representations of the model to work backwards. - Rearrange the high tide equation to find which day of the month has the high tide falling at 12 hours after midday. - Derive an equivalent model for the low tide. - Recognises limitations of the model when moving from the end of one month to another and suggests a solution. Border problem: - Manipulate the range of algebraic representations to demonstrate that they are all equivalent ie a) $4n - 4$ b) $2n + 2(n - 2)$ c) $4(n - 2) + 4$ d) $4(n - 1)$ - Determine if any number of tiles will border a square.	solution back in the real-world context d) justification of the mathematical representations e) interpreting, reviewing, and generalising the model in context. - Extend models to generalise solutions. - Eg determines a month-by-month solution for the high and low tides times. - Border problem – derive a model for any $m \times n$ rectangle and use this to determine which rectangles could be bordered by any number of tiles.

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	With support, students make and test conjectures involving linear relations using digital tools. For example, students can: • Test a given conjecture relating to the gradient or intercept of a linear relation using digital tools. • Eg Bek has come up with the conjecture “If the coefficient of x is negative, the graph will slope down from left to right.” a) Test Bek’s conjecture and conclude whether or not her conjecture is correct.	Students make and test conjectures involving linear relations using digital tools. For example, students can: • Make conjectures relating to either the gradient or intercept of linear relations (linear functions and inequalities) in gradient-intercept form. Using graphing software, tests the conjectures to support or refute them. • Eg using graphing software, such as Desmos, tests multiple conjectures relating to either the slope OR the intercept. Such as how the sign affects the slope and how the magnitude affects the slope. • For inequalities, contrast graphs of $y = 2x$ with related inequalities (see the content elaboration F-10 Curriculum Mathematics – Year 8 for an exhaustive list), making and testing conjectures about sign and the direction of the inequality.	Students make and test multiple conjectures involving linear relations using digital tools. For example, students can: • Make conjectures relating to linear relations in gradient-intercept form. Using graphing software, tests the conjectures to support or refute them. • Eg use graphing software, such as Desmos, test multiple conjectures regarding changes to the coefficient and constant of linear relations, including related inequalities, such as how the sign and magnitude affect both the slope and intercept, and the direction of the inequality affects the outcome.	Students make and test multiple conjectures involving linear relations other than slope-intercept form, using digital tools. For example, students can: • Make conjectures relating to linear relations in a form other than gradient-intercept form. Using graphing software, tests the conjectures to support or refute them. • Eg using graphing software, such as Desmos, predicts, tests, and summarises integer solutions to the linear inequality: $2x + 3y \leq 12$.

Measurement

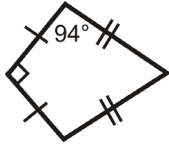
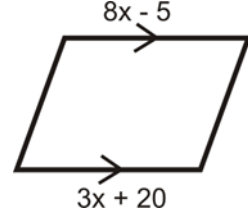
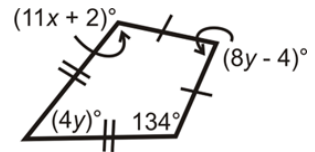
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Measurement	With support, students use appropriate metric units when solving measurement problems involving the perimeter and area of composite shapes, and volume of right prisms. For example, students can: - Use metric units to find perimeters of 2D shapes. - Use formulas to find the volume of right prisms, eg, if given a cross-sectional area and height, calculate the volume of a right prism.	Students use appropriate metric units when solving measurement problems involving the perimeter and area of composite shapes, and volume of right prisms. For example, students can: - Use models to demonstrate the number of cubic centimetres in a cubic metre and relating this to capacities of millilitres and litres. - Solve practical problems involving volume and capacity; for example, optimal packaging and production. - Find area and/or perimeter of composite shapes, made up of to three polygons. eg, using arrays and rectangles to approximate the area of irregular shapes in situations such as a farmer needing to work out how much seed, fertilizer and herbicide are required to cover a paddock. - Eg find the area of this composite shape. 	Students use appropriate metric units when solving complex measurement problems involving the perimeter and area of composite shapes, and volume of right prisms, other than those with rectangular and triangular cross-sections. For example, students can: - Solve complex practical problems involving perimeter and area, eg, can identify the minimum information needed to find the perimeter and area of a composite shape such as:  - Find a missing dimensions in right prisms other than rectangular or triangular prisms (ie, regular and irregular polygons, in problems that emphasise that the volume = cross-sectional area $\times$ height relationship) eg if the volume of a typical honey cell in a beehive is 350 mm^3 and the cells are 18 mm deep, what is the cross-sectional-area of the cell? Provide reasonable estimates for the side lengths of the hexagon.	Students use appropriate metric units when solving complex measurement problems involving the perimeter and area of composite shapes, and volume of right prisms that involve composite shape cross-sections. For example, students can: - Solve complex composite shape problems such as Inaba Area Mazes:  - Can solve multistep problems involving right prisms with cross-sections that are composite shapes in problems that emphasise that the volume = cross-sectional area $\times$ height relationship) eg  The volume of this case (a right prism) is 360 cm^3, and the height is 45 mm. What is the

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		- Connect volume and capacity and explain the difference between the two. - Determine an unknown dimension if given the volume and 2 other dimensions or find the cross-sectional area if given a height and volume of rectangular and triangular right prisms.		cross-sectional area? Suggest and test reasonable dimensions for the lid.
	With support, students use Pythagoras' theorem to solve measurement problems involving unknown lengths of right-angle triangles. For example, students can: - Locate and name the hypotenuse. - Demonstrate Pythagoras' theorem using concrete materials or virtual manipulatives. Eg Geogebra.	Students use Pythagoras' theorem to solve measurement problems involving unknown lengths of right-angle triangles. For example, students can: - Use Pythagoras' theorem to determine unknown side lengths of right-angled triangles in practical applications (eg length of ladder/height of wall, construction examples, 2D navigation - taking the shortest path). - Use Pythagoras' theorem to determine where a triangle is right or otherwise.	Students use Pythagoras' theorem to solve multi-step measurement problems involving unknown lengths of right-angle triangles. For example, students can: - Use Pythagoras' theorem to determine where a triangle is right or otherwise. - Use Pythagoras' theorem to solve multi-step problems, eg 2 pieces of timber are used to support the wall of a building during construction. What is the length of the shorter piece? 	Students use Pythagoras' theorem to solve complex and multi-step measurement problems involving unknown lengths of right-angle triangles. - For example, students can: - Use Pythagoras' theorem to solve multiple step problems that may require the students to create right triangles and may be in three dimensions. Eg this bridge cross-section is symmetrical, find the length of the side marked x. - Eg find length AE of this square based pyramid, where 'x' is in the centre of the square base.  

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	With support, students use formulas to solve problems involving the area and circumference of circles. For example, students can: - Use supported steps of a given formula to solve problems involving the area and circumference of circles. - Eg use these steps to help solve: $A = \pi \times 6 \times 6$ 	Students use formulas to solve problems involving the area and circumference of circles. For example, students can: Given any of the radius, diameter, circumference, or area of a circle, deduce the others.	Students use formulas to solve composite shape problems involving the area and circumference of circles. For example, students can: Use circumference and area formulas to solve sector and arc problems, including composite shapes: a) Find the area and perimeter of this shape. 	Students use formulas to solve problems involving the area and circumference of circles, including working backwards in arc and sector problems. For example, students can: Use circumference and area formulas to solve arcs and sector problems that involve finding the radius or angle, such as: a) When given arc length and angle determine the radius, or if given the sector area and radius, determine the angle.
	With support, students solve problems of duration involving 12- and 24-hour cycles across multiple time zones. For example, students can: - Recognise time written in 12- and 24-hour notation. - Recognise that time-zones relate to the lines of longitude of Earth.	Students solve problems of duration involving 12- and 24-hour cycles across multiple time zones. For example, students can: - Convert between 12- and 24-hour time representations. - Determine the local time when arriving in another time zone from an international flight, eg Sydney to Dubai. - Plan a 2hr meeting with another school in Amsterdam. Explain and justify when you should meet to be most convenient for both parties.	Students solve multi-step problems of duration involving 12- and 24-hour cycles across multiple time zones, involving working from. For example, students can: Given departure and arrival times for journeys across multiple time zones, determine the duration of the journey.	

Space

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Space	With support, students use 3 dimensions to locate and describe position. For example, students can: Given a 3D shape, describe the vertices of the shape using positional language.	Students use 3 dimensions to locate and describe position. For example, students can: - Create 3D shapes using 3D cartesian coordinates in the first octant and identify the vertices of the shape with positional triples (x, y, z), with or without digital tools. - Use 3D dimensions such as latitude, longitude and altitude to describe the positions of aircraft at a point in time. - Be given a multilevel car park, locate the position of empty car parks and give directions for a person to find the empty car parks from the car park entrance.		
	With support, students identify conditions for congruency and similarity in shapes and create and test algorithms designed to test for congruency and similarity. For example, students can: - Recognise some shapes that are similar. - Identify 2 triangles as congruent if one lies exactly over the top of another. - Test a given algorithm (eg flowchart) to test for congruency or similarity of sets of triangles.	Students identify conditions for congruency and similarity in shapes and create and test algorithms designed to test for congruency and similarity. For example, students can: - Use the congruency tests (SSS, SAS, ASA, AAS and HL) to determine congruency of or otherwise of triangles. - Use the similarity tests (AA, SSS – sides in proportion, and SAS – sides in proportion). - Create and test algorithms (eg flowcharts) that can be used to determine the congruency and similarity of triangles.	Students identify conditions for congruency and similarity in shapes and create test and evaluate algorithms designed to test for congruency and similarity. For example, students can: Analyse given algorithms (eg flowcharts designed by other students) designed to determine congruency and similarity of triangles and quadrilaterals and suggest improvements.	

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	With support, students apply the properties of quadrilaterals to solve problems. For example, students can: Recognise some properties of common quadrilaterals.	Students apply the properties of quadrilaterals to solve problems. For example, students can: - Distinguish between different quadrilaterals using their properties. - Use the properties of quadrilaterals to solve problems including a missing side length or angle.	Students apply the properties of quadrilaterals to solve problems to find more than one angle or side length. For example, students can: - Use the properties of quadrilaterals to solve multi-step problems. - Eg Find the sizes of the other angles:  - A parallelogram has two angles of 46°, what are the other two angles?	Students apply the properties of quadrilaterals to solve algebraic problems. For example, students can: - Solve problems using algebraic side lengths. - Eg find the value of x for this quadrilateral to be a parallelogram:  - Find the values of the missing variables: 

Statistics

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Statistics	With support, students conduct statistical investigations and explain the implications of obtaining data through sampling. For example, students can: - Undertake statistical investigations involving samples of a population, identifying the subjects and variables to be investigated, planning the approach, collecting data, representing, interpreting and analysing the data, making inferences and communicating the results of the investigation in terms of the context, acknowledging uncertainty of the chosen sampling method. - Conduct a statistical investigation through sampling and can construct a table or graph display.	Students conduct statistical investigations and explain the implications of obtaining data through sampling. For example, students can: Undertake statistical investigations involving samples of a population, identifying the subjects and variables to be investigated, planning the approach, collecting data, representing, interpreting and analysing the data, making inferences and communicating the results of the investigation in terms of the context, acknowledging the implications and uncertainty of the chosen sampling method.	Students conduct statistical investigations and explain and compare the implications of obtaining data through different sampling methods. For example, students can: - Undertake statistical investigations involving samples of a population, identifying the subjects and variables to be investigated, planning the approach, collecting data, representing, interpreting and analysing the data, making inferences and communicating the results of the investigation in terms of the context, acknowledging the implications and uncertainty of the chosen sampling method. - Compare and explain the implications of different sampling methods.	Students conduct statistical investigations and explain, compare and evaluate the implications of obtaining data through sampling. For example, students can: - Undertake statistical investigations involving samples of a population, identifying the subjects and variables to be investigated, planning the approach, collecting data, representing, interpreting and analysing the data, making inferences and communicating the results of the investigation in terms of the context, acknowledging the implications and uncertainty of the chosen sampling method. - Compare and evaluate the implications of different sampling methods.
	With support, students analyse and describe the distribution of data. For example, students can: Use a random or non-random sampling technique to select samples, and describe the shape of the data distribution.	Students analyse and describe the distribution of data. For example, students can: - Use random and non-random sampling techniques to select samples from primary and secondary sources. - Analyse and describe the distribution of data using terms such as positive skew, negative skew, symmetric and bi-modal.	Students analyse, describe and compare the distribution of data sets. For example, students can: Make comparisons between different data sets, generated by different sampling techniques from the same primary or secondary sources, and can explain these in terms of distribution.	Students analyse, describe and compare the distribution of data sets, identifying and justifying which sampling techniques are appropriate for particular contexts. For example, students can: - Justify, in terms of data distribution, which sampling techniques are appropriate for different contexts. - Eg may repeat the same sampling technique on a data set, and justify if it is appropriate or not, on the basis of changes to the distribution with every trial.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	With support, students compare the variation in distributions of random samples of the same and different size from a given population with respect to shape, measures of central tendency and range. For example, students can: Compare two different random samples from the same population and make some comparisons between the samples.	Students compare the variation in distributions of random samples of the same and different size from a given population with respect to shape, measures of central tendency and range. For example, students can: - Make repeated samples of the same and different sizes of the same population, comparing each in terms of shape, range and measures of central tendency. - Recognise that larger samples will tend to be more representative.	Students compare the variation in distributions of random samples of the same and different size from a given population with respect to shape, measures of central tendency and range and makes reasoned conclusions about the minimum sample size of the population. For example, students can: Draw reasonable conclusions about the minimum size of random sample sizes required to estimate the characteristics of a population.	

Probability

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Probability	With support, students represent the possible combinations of 2 events with tables and diagrams, and determine related probabilities to solve practical problems. For example, students can: - Represents the possible combinations of 2 events with either a table or a diagram. - Eg Displaying the sample space for a spinner and die using a two-way table or a diagram.	Students represent the possible combinations of 2 events with tables and diagrams, and determine related probabilities to solve practical problems. For example, students can: - Represent probabilities using a 2-way table, tree diagram or Venn Diagram to help visualise the probabilities of two events such as an equal sector 4-part spinner and rolling a standard die. - Recognise that complementary events have a combined probability of one, and use this in the case of 2 events, to develop and use the relationship: $\Pr(A \text{ and } B) + \Pr(A \text{ and not } B) + \Pr(\text{not } A \text{ and } B) + \Pr(\text{not } A \text{ and not } B) = 1$	Students represent the possible combinations of 2 events with tables and diagrams, and determine related probabilities to solve practical problems for events with unequal likelihoods. For example, students can: Determine related probabilities to solve practical problems involving for example, with non-standard dice and spinners with unequal sectors.	Students represent the possible combinations of 2 events with tables and diagrams, and determine related probabilities to solve practical problems for events, working backwards from a stated probability. For example, students can: - Given a probability of 2 events, determine the probability of each individual event. - Eg design two 10-sided dice so the possibility of rolling a total of 8 is 0.6 when they are both rolled.
	With support, students conduct experiments and simulations using digital tools to determine related probabilities of compound events. For example, students can: Conduct either an experiment or a simulation and determine some probabilities of compound events.	Students conduct experiments and simulations using digital tools to determine related probabilities of compound events. For example, students can: - Roll two 6-sided dice 60 times, then use an online simulator (such as Dice Roll Simulation – GeoGebra) and compare the results for 600, 6 000 and 60 000 rolls (related probabilities derived from are multiple trials of experiments). - Describes difference being related to the number of trials, that is, that theoretical probabilities for seemingly random events will only be approached if the number of trials is large enough.	Students design and conduct experiments and simulations using digital tools to determine related probabilities of compound events. For example, students can: For non-standard compound events as above (non-standard die and spinners with unequal sectors) run an experiment 50 times, and design a matching simulation (eg using random number generators in Excel) and run many times to compare calculated and related probabilities, and explain any differences between them.	Students design and conduct experiments and simulations using digital tools to determine related probabilities of compound events, working backwards from a stated probability. For example, students can: Given a problem that involves working backwards from a probability for 2 events such as “design two 10-sided dice so the possibility of rolling a total of 8 is 0.6 when they are both rolled”, design a matching simulation using the student’s suggested dice, to determine the related probabilities, to compare calculated and related probabilities, explain any differences between them and refined the solution if necessary.

